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(57) Abstract :

ABSTRACT HAZARDOUS GAS DETECTION KIT USING RF COMMUNICATION FOR SEWAGE WORKERS The present invention discloses a gas-detecting wearable belt designed for sewage cleaning operations. Comprising a set of sensors (1-4), transmitter (7) and receiver, it detects toxic gases such as hydrogen sulphide, ammonia, methane, carbon monoxide, and Sulphur dioxide. Upon detection, it triggers a buzzer (6) and illuminates a red LED (9) for wearer caution. A pulse sensor (5) monitors heart rate, with automated mechanisms activating if gas levels rise or heart rate decreases significantly. In emergencies, a motor (10) pulls the wearer out of the drainage system, enhancing safety. An external buzzer (14) alerts attendants, minimizing response time in hazardous situations. This innovative dual-alert system prioritizes sewage cleaner well-being, ensuring swift responses and risk mitigation. Ref fig. 1 & 2

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